MIL-P-20088C
14 October 1986
SUPERSEDING
MIL-P-20088B
21 November 1952
(See 6.7)

MILITARY SPECIFICATION

PRESERVATION SOLUTION CONCENTRATE, SONAR SEA CHEST

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers preservation solution concentrate for use as a filling solution in sonar sea chests. This material may be used in geographical areas where air pollution regulations exist (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

O-M-232 - Methanol (Methyl Alcohol).
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon (208 Liters).
PPP-P-704 - Pails, Metal: (Shipping, Steel, 1 Through 12 Gallons).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.
FED-STD-313 - Material Safety Data Sheets Preparation and the Submission of.
FED-STD-791 - Lubricants, Liquid Fuels, and Related Products; Methods of Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-290 - Packaging of Petroleum and Related Products.

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 562 - Standard Test Method for Consistency of Paints Using the Stormer Viscometer. (DoD adopted)
D 1475 - Test Method for Density of Paint, Varnish, Lacquer, and Related Products. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rules and Regulations - Rule 102

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 E. Flair Drive, El Monte, CA 91731.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Composition. The composition, by weight, of the preservation solution concentrate shall be as specified in table I.

TABLE I. Composition.

Component	Percent by weight	
	Minimum	Maximum
Ethylene glycol[1]	75.0	---
Sodium tetraborate ($Na_2B_4O_7$)	2.5	2.7
Disodium phosphate	0.45	0.55
Monosodium phosphate	0.45	0.55
Sodium mercaptobenzothiazole[2]	0.45	0.55
Water (distilled)	---	22.0

[1]Ethylene glycol shall be specified as follows:

	Minimum	Maximum
Specific gravity at 20/20 degrees Celsius (°C)	1.1151	1.1156
Boiling range at 760 millimeter (mm)	190°C	210°C
Acidity as acetic acid, percent by weight	----	0.010

[2]Sodium mercaptobenzothiazole shall be as specified in 3.1.1.

3.1.1 Sodium mercaptobenzothiazole. The sodium mercaptobenzothiazole shall be added in the form of an aqueous solution consisting essentially of sodium mercaptobenzothiazole and water. The aqueous solution shall contain not less than 45 percent by weight of sodium mercaptobenzothiazole, determined as specified in 4.6.2, and shall contain not more than 1 percent by weight of chemically pure sodium sulfite (Na_2SO_3), as a stabilizer. The aqueous solution shall be known as "NaMBT solution".

3.1.1.1 "NaMBT solution". The "NaMBT solution" shall conform to the following requirements.

3.1.1.1.1 Specify gravity. The specific gravity at 25/25°C (77/77 degrees Fahrenheit (°F)) shall be 1.27 ± 0.03.

3.1.1.1.2 Color. The color shall be not darker than No. 15 in the Gardner Color Standards of 1933, determined as specified in 4.6.4.

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3.1.1.1.3 Clarity. The "NaMBT solution" shall be free from suspended matter.

3.1.1.1.4 Solubility in water. When diluted with distilled water to form a solution containing 0.2 percent by weight of sodium mercaptobenzothiazole, not more than a slight haze shall be formed after standing 24 hours at 25°C (77°F).

3.1.1.1.5 Solubility in ethylene glycol. The "NaMBT solution" shall be completely soluble in the preservation solution concentrate when added in the ratio of three parts by weight of "NaMBT solution" to 100 parts by weight of the concentrate. The final mixture shall not develop any haze or cloudiness when after being thoroughly mixed, it is allowed to stand undisturbed and protected from direct sunlight, at a temperature of 25 ± 5°C (77 ± 8°F) for 24 hours.

3.1.1.1.6 Free alkalinity. The free alkalinity expressed as NaOH shall be 0.05 to 0.75 gram per 100 milliliters (mL) at 25 ± 3°C (77 ± 5°F) when determined as specified in 4.6.5.

3.2 pH value. The pH value of a mixture of one part preservation solution concentrate and four parts distilled water (by volume) shall be 7.8 to 8.2 at 25°C (77°F), determined as specified in 4.6.7.

3.3 Boiling point. The boiling point of the preservation solution concentrate shall be not less than 118°C (245°F) at one atmosphere pressure. The boiling point of one part preservation solution concentrate and four parts of distilled water (by volume) shall be not less than 101°C (214°F) at one atmosphere pressure. The boiling points shall be determined as specified in 4.6.8.

3.4 Ash. The ash content of the preservation solution concentrate shall not exceed 5 percent by weight determined as specified in 4.6.9.

3.5 Density. The density at 20°C (68°F) of a mixture of one part preservation solution concentrate and four parts distilled water (by volume) shall be at least 1.02 grams cubic centimeters (g/cc).

3.6 Viscosity. The viscosity of a mixture of one part preservation solution concentrate and four parts distilled water (by volume) shall be 0.74 ± 0.03 centistokes at 54°C.

3.7 Solvent. The solvent portion of the solution shall be in accordance with the following requirements:

(a) A combination of hydrocarbons, alcohols, aldehydes, ethers, or ketones having an olefinic or cycloolefinic type of unsaturation except perchloroethylene: 5 percent maximum.
(b) A combination of aromatic compounds with eight or more carbon atoms to the molecule except ethylbenzene, methyl benzoate, and phenyl acetate: 8 percent maximum.
(c) A combination of ethylbenzene, ketones having branched hydrocarbon structures, trichloroethylene, or toluene: 20 percent maximum.

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Solvents shall not exceed 20 percent by volume of the solution. The volatile thinner shall be in accordance with any solvent system in accordance with Rule 102 of the South Coast Air Quality Management District (see 4.9).

3.8 Hazardous materials. Lead, chromate, or asbestos shall not be present.

3.9 Toxicity. The solution shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting activity.

3.10 Material safety data sheet. The contracting activity shall be provided a material safety data sheet (MSDS) at the time of contract award. The MSDS form and requirements for its preparation are found as specified in FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification (see 6.5).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Quality conformance inspection.

4.2.1 Lot. For purposes of sampling, a lot shall consist of all material manufactured as one batch, and offered for delivery at one time.

4.2.2 Sampling for inspection of filled containers. A random sample of filled containers shall be selected from each lot by the Government inspector in accordance with MIL-STD-105 at inspection level I and Acceptable Quality Level (AQL) of 2.5 percent defective, to verify compliance with all stipulations of this specification not involving tests.

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4.2.3 Sampling for quality conformance inspection. From each inspection lot, the Government inspector shall select five containers at random. From each of the five containers 1-quart specimens shall be taken and placed in separate clean, dry, metal or glass containers, sealed, marked, and forwarded to the testing laboratory designated by the Naval Sea Systems Command.

4.3 Inspection procedures.

4.3.1 Inspection of filled containers. Each sample filled container selected as specified in 4.2.2 shall be examined by the Government inspector for defects of the container and the closure, for evidence of leakage, and for unsatisfactory markings; each sample filled container shall also be weighed to determine the amount of the contents. Any container in the sample, having one or more defects, or under required fill, shall be rejected and if the number of defective containers in any sample exceeds the acceptable number for the appropriate sampling plan in accordance with MIL-STD-105, the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for acceptance inspection provided the contractor has removed or repaired all nonconforming containers.

4.4 Tests. The sample specimens selected as specified in 4.2.3 shall be subjected separately to the tests as specified in 4.6. If any specimen fails one or more of these tests, the lot shall be rejected. Rejected lots may be resubmitted for acceptance tests provided the contractor has removed or reworked all nonconforming products.

4.5 Tests of ingredient materials.

4.5.1 When specified by the bureau or agency concerned, representative samples of materials used in the manufacture of the preservation solution concentrate shall be selected for test by the Government inspector.

4.5.2 Tests on ingredient materials, when specified, shall consist of all the tests as specified in 4.6, and shall be conducted at the place of manufacture, under the supervision of the Government inspector. When facilities are not available at the place of manufacture, tests shall be conducted at a laboratory designated by the bureau or agency concerned.

4.5.3 Test reports shall be prepared in accordance with the data ordering documents included in the contract or order (see 6.2.2).

4.6 Test methods. Appropriate tests shall be employed as are necessary to determine conformance with this specification, in addition to the following:

4.6.1 Phosphate content. Testing for phosphate content shall be conducted as follows:

4.6.1.1 Preparation of magnesia mixture. Dissolve 50 grams of crystalline magnesium chloride ($MgCl_2 \cdot 6H_2O$) and 100 grams of ammonium chloride in 500 mL of water. To this solution add about 10 mL of concentrated aqueous ammonia, allow to stand overnight, and filter if a precipitate appears. Make this solution slightly acid with hydrochloric acid, dilute to 1 liter (L), and keep in a glass stoppered bottle.

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4.6.1.2 Procedure. Dilute a 20-gram sample, weighed to the nearest 0.01 gram of the preservation solution concentrate with 100 mL of distilled water. To this mixture add 10 mL of a mixture of equal parts by volume of concentrated hydrochloric acid and water, and 30 mL of magnesia mixture. Add concentrated aqueous ammonia until the solution is neutral to methyl red indicator. Stir for 5 minutes, and while continuing to stir, add 15 mL additional aqueous ammonia. Allow the solution to stand for 4 hours or longer, filter, and ignite the precipitate in a weighed platinum crucible at 1000 to 1050°C (1832 to 1922°F) to constant weight. Calculate the phosphate concentration from the weight of the magnesium pyro-phosphate ($Mg_2P_2O_7$) as follows:

$$\text{percent phosphate} = \frac{\text{weight of } Mg_2P_2O_7 \times 0.85337 \times 100}{\text{weight of sample}}$$

4.6.2 Sodium mercaptobenzothiazole.

4.6.2.1 Sodium mercaptobenzothiazole content of the finished preservation solution concentrate. The sodium mercaptobenzothiazole content of the finished preservation shall be determined as follows:

4.6.2.2 Weigh out accurately (to the nearest 0.01 gram) a 10.00 to 12.00-gram sample of the finished preservation solution concentrate into a clean, tared 250 mL glass stoppered Erlenmeyer flask. If sample is turbid, it should be filtered before weighing.

4.6.2.3 Add 10 mL of methanol in accordance with grade A of O-M-232, and mix the liquids by shaking the flask, (to aid in keeping the "NaMBT" in solution).

4.6.2.4 Add diluted (approximately 3 normal (N)) hydrochloric acid dropwise, shaking the flask frequently to mix the contents, until the pH value of the solution is 4.0 ± 0.2, as indicated by appropriate indicator paper or by a pH meter.

4.6.2.5 Add exactly 20 mL of standardized N/50 iodine solution, and mix well by shaking the flask. Adjust the temperature of the mixture to 25 ± 5°C (77 ± 8°F), and allow the flask to stand at this temperature for approximately 15 minutes to insure completion of the reaction of iodine with "NaMBT".

4.6.2.6 Add approximately 100 mL of distilled water; mix well by shaking the flask, and adjust the temperature of the mixture to 25 ± 5°C (77 ± 8°F).

4.6.2.7 Titrate the excess (unconsumed) iodine with standardized N/50 sodium thiosulfate solution, using starch as an indicator.

4.6.2.8 (mL of I_2 x normality) - (mL of $Na_2S_2O_3$ x normality) = mL of N/1 iodine consumed by sample.

$$\frac{\text{mL of N/1 } I_2 \text{ consumed} \times 0.1892 \times 100}{\text{weight of sample, in grams}} = \text{percent "NaMBT" by weight}$$

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4.6.3 Sodium mercaptobenzothiazole content of "NaMBT solution". The sodium mercaptobenzothiazole content of the "NaMBT solution" shall be determined as follows:

4.6.3.1 Weigh out accurately (to the nearest 5 milligrams (mg)) a 5.000 to 5.200 gram sample of "NaMBT" into a clean, tared 250 mL glass beaker.

4.6.3.2 Add approximately 100 mL of distilled water. Add chemically pure (cp) concentrated hydrochloric acid cautiously, with constant stirring, until the solution is just acid to litmus and then add 10 drops of the 12N hydrochloric acid in excess. Stir the mixture vigorously with a glass stirring rod for approximately 1 minute. Cool the mixture to a temperature of 10 to 15°C (50 to 59°F) and allow it to stand at this temperature for 1 hour.

4.6.3.3 Prepare a filter consisting of a Gooch crucible and mat of asbestos or equivalent and dry it in an oven at 105 to 110°C (221 to 230°F), to constant weight. Filter the mixture through the tared Gooch crucible, using suction. Wash the precipitate in the crucible with five 10 mL portions of cold distilled water, allowing each portion of wash water to pass through the filter before adding the next portion. Remove as much of the last portion of wash water as possible by suction.

4.6.3.4 Dry the precipitate (in the crucible) for 3 hours in an oven at 105 to 110°C (221 to 230°F). Cool the crucible and contents for at least 15 minutes in a desiccator, and weigh. Repeat the drying and weighing to constant weight.

$$\frac{\text{weight in grams of precipitate}}{\text{weight in grams of sample}} \times 1.1316 \times 100 = \text{percent "NaMBT" by weight}$$

4.6.4 Color of the "NaMBT solution". A representative sample of the aqueous sodium mercaptobenzothiazole solution used as an inhibitor shall be examined visually in a Hellige Color Comparator, or equivalent. The result shall be expressed as a number in the scale of the Gardner Color Standards of 1933 used by the Institute of Paint and Varnish Research. An approximate conversion of color designation from one system to that of any other system can be accomplished by means of method 4241 of FED-STD-141.

4.6.5 Free alkalinity of the "NaMBT solution". The free alkalinity shall be determined as follows:

4.6.5.1 Using a 10 mL transfer pipette, 10 mL of the sample at 25 ± 3°C (77 ± 5°F) shall be transferred to a clean 250 mL glass beaker. Precautions should be taken to obtain only the clear solution for analysis.

4.6.5.2 Prepare a mixture consisting of equal parts by volume of uninhibited ethylene glycol and distilled water. Add 100 mL of the glycol-water mixture to the sample of "NaMBT" in the beaker. Volume measurements may be made with a graduate cylinder.

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4.6.5.3 Titrate the solution potentiometrically, using a pH meter equipped with a glass electrode and a suitable reference electrode, to a pH value of 9.0 ± 0.05, using standardized N/10 or N/20 hydrochloric acid. During the titration, the solution shall be stirred effectively. A glass stirrer, driven by a small electric motor or by compressed air may be used. Stirring shall not be accomplished by blowing air through the solution as this tends to oxidize the "NaMBT".

$$\frac{\text{mL of HCl} \times \text{normality of HCL} \times 0.040 \times 100}{\text{mL of sample}} = \text{grams of NaOH per 100 mL of sample}$$

4.6.6 Water content. The water content shall be determined in accordance with method 3001.9 of FED-STD-791, except that pure benzene instead of naphtha shall be employed and the weight of water contained in the sample shall be calculated.

4.6.7 pH value. The pH value shall be determined potentiometrically using a pH meter equipped with a glass electrode and a suitable reference electrode (see 3.2).

4.6.8 Boiling point. One hundred mL of the mixture as specified in 3.3, together with a few chips of unglazed porcelain or other suitable material to prevent superheating, shall be placed in a 200 mL round bottom flask fitted with a reflux condenser. The inner tube of the condenser shall be wide enough to accommodate a thermometer safely. A thermometer shall be passed through the reflux condenser by means of a long, thin wire in such a way that the bulb of the thermometer shall be completely immersed in the liquid. No part of the bulb of the thermometer shall be closer than 1/2 inch to the bottom of the flask, and no part of the thermometer shall be in contact with the flask. The liquid mixture shall be heated on a hot plate with refluxing until the equilibrium temperature is reached. At this point the contents of the flask shall be swirled around in such a way that the moisture condensed on the walls of the flask shall be washed down into the body of the liquid. The temperature at which the liquid boils immediately after this treatment shall be taken as the boiling point of the product.

4.6.9 Ash. The ash content of the preservation solution concentrate shall be determined by evaporation of the major portion of the glycol and ignition of the residue in accordance with the following procedure: Ignite a clean size 1A glazed porcelain or Vycor crucible containing approximately 1.0 gram of analytical grade shredded asbestos or equivalent in a muffle furnace for 1 hour at a temperature of 875 ± 15°C (1607 ± 27°F). Cool the crucible and contents for 10 to 15 minutes in the open air, then for 15 to 20 minutes in a desiccator, and weigh on an analytical balance. Add from 5 to 7 grams of the ethylene glycol to the crucible; and immediately weigh the crucible and contents as rapidly as possible on an analytical balance. Evaporate the gylcol for a period of 4 hours at a temperature of 160 to 190°C (320 to 374°F) on an electric hot plate or by means of an 'Infraradiator' or comparable apparatus. Control the temperature so that there is no loss of material by spattering during the evaporation. Next,

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place the crucible in a muffle furnace at a temperature not higher than 200°C (392°F); and gradually raise the temperature to 875 ± 15°C (1607 + 27°F) over a period of 2 hours. Maintain the temperature at 875 ± 15°C (1607 + 27°F) for a period of 1 hour. Cool the crucible and contained ash for 10 to 15 minutes in the open air, then for 15 to 20 minutes in a desiccator; and weigh rapidly on an analytical balance. Calculate the percent of ash as follows:

$$\frac{\text{Weight of ash}}{\text{Weight of sample}} \times 100 = \text{percent of ash}$$

(The asbestos or equivalent is used to prevent loss of sample by spattering during evaporation and ignition. Satisfactory results cannot be obtained by direct ignition of glycol over an open flame. Rapid weighing is required because of the hygroscopic nature of glycol and of the ash.)

4.6.10 Density. The density tests shall be in accordance with ASTM D 1475.

4.6.11 Viscosity. The viscosity tests shall be in accordance with ASTM D 562.

4.7 Toxicity. The contractor shall furnish to the contracting activity the toxicological data and formulations required to evaluate the safety of the material for the proposed use.

4.8 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

4.9 Certification data/report. When specified in the contract or order, each lot of solution shall be certified (see 6.2.2).

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Preservation-packaging. The preservation solution concentrate shall be delivered in 5-gallon closed top steel containers in accordance with PPP-C-96 or PPP-P-704 or in 55-gallon drums made from 0.0478 ± 0.005 inch steel (18 gauge) in accordance with PPP-D-729. The size of container desired shall be as specified (see 6.2.1).

5.1.1 Preservation solution concentrate shall be preserved-packaged and packed for the level of protection specified in accordance with MIL-STD-290 (6.2.1).

5.2 Packing. Five-gallon closed top steel pails and 55-gallon drums will require no further packing.

5.3 Marking. In addition to any special marking required by the contract or order, containers shall be marked and labelled in accordance with MIL-STD-129 and with the following information:

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STORAGE SOLUTION CONCENTRATE
Specification Number

POISON! - THIS MATERIAL IS POISONOUS IF TAKEN INTERNALLY AND MAY PROVE FATAL

QUANTITY _____
CONTRACTOR _____
MANUFACTURER _____
CONTRACT NO. _____
BATCH NO. _____

5.3.1 Special marking. In addition to other markings required on the containers, there shall be the following statement: "The volatile content of this container is not photochemically reactive as defined by rule 102 of the South Coast Air Quality Management District (California)" (see 6.2.1 and 6.3).

6. NOTES

6.1 Intended use. The preservation solution concentrate specified herein is intended to be used as a filling solution in sonar sea chests.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of container required (see 5.1).
(c) Level of protection of preservation-packaging and packing required (see 5.1.1).
(d) Special marking, when required (see 5.3.1).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.5.3	Reports, test	DI-T-2072	----
4.9	Certification data/report	UDI-A-23264	10.2.1

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., Vol. I, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Volatile content. Although the container marking specifically refers to the South Coast Air Quality Management District (California), the solution may be used anywhere material complying with 3.7 are allowed. This includes nearly all other air pollution control districts or similar areas controlling the emission of solvents into the atmosphere. Information concerning rule 102 may be obtained from the South Coast Air Quality Management District, 9150 E. Flair Drive, El Monte, CA 91731.

6.4 Subject term (key word) listing.

Preservation solution
Sonar
Sea chest

6.5 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in appendix B of FED-STD-313.

6.6 In addition to the requirements specified in 4.9, the certification data/report should include the following information:

(a) Toxicological data and formulations required to evaluate the safety of the material for the proposed use (see 3.9).
(b) Certification that volatility requirements are met (see 3.7).
(c) Conformance of each lot of ingredient material, stating test results and source, as applicable (see 4.6).

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Navy - SH

Review activities:
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Preparing activity:
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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-20088C	PRESERVATION SOLUTION CONCENTRATE, SONAR SEA CHEST

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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